Exploring Aligned and Disaligned Multimodal Features in Recommender Systems

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January 19, 2026

Outline

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Recommender Systems

- Guide users to relevant items in a large space [5]
- Approaches: Content-Based, Collaborative, Hybrid, Knowledge-Based, Context-Aware

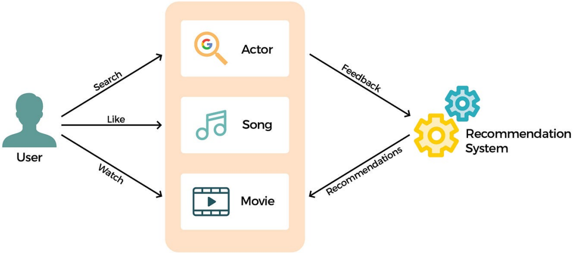


Figure: Example of a Recommender System

Multimodal Recommender Systems

- Use multiple modalities (text, image, audio) to enrich item representations [7]
- Encoding options: concat, fusion/attention, GNNs [1, 4]

Figure: Example of multimodal data for a movie (Star Wars)

Motivation & Problem Statement

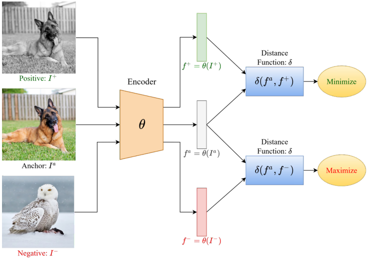
Traditionally, MMRec systems extract features using independent encoders (e.g., BERT for text, ResNet for images). In other domains, like Few-Shot Learning for Image Classification, aligned features (CLIP) have shown superior performance with respect to disaligned ones.

Outline

Contrastive Learning

- Self-supervised method: learns representations that bring similar samples closer [3]
- **Goal:** Minimize distance between positive pairs, maximize distance between negative pairs.
- Contrastive loss:

$$\ell_{i,j} = -\log \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} [k \neq i] \exp(\mathrm{sim}(z_i, z_k)/\tau)}$$



CLIP (Contrastive Language-Image Pre-training)

- Aligns image and text embeddings in a shared space (OpenAI) [6]
- **Relevance:** Large multimodal encoders like CLIP can improve performance when fine-tuned end-to-end [8]

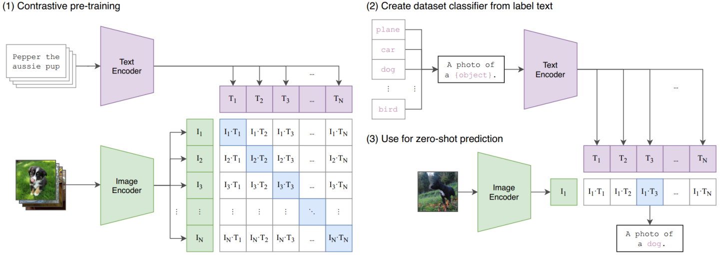


Figure: CLIP aligns visual and textual concepts

AudioCLIP

- Extends CLIP by adding an **audio encoder** (ESResNeXt) [2]
- **Training process:**
  1. Audio-head pre-training
  2. Cooperative distillation (aligning audio to frozen CLIP space)
  3. Full tri-modal fine-tuning on AudioSet

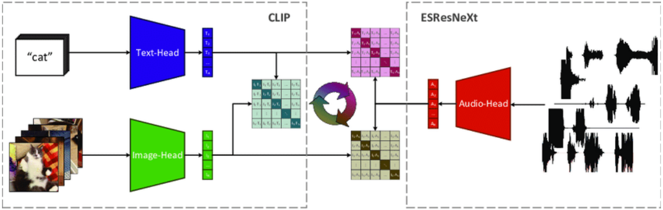


Figure: Tri-modal alignment in AudioCLIP

Outline

Research Questions

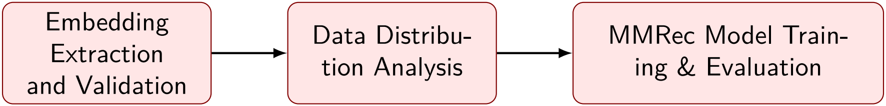
RQ1: Qualitative Analysis (Geometry)

How does the alignment process (e.g., CLIP, AudioCLIP) affect the geometric distribution of the feature space?

RQ2: Quantitative Analysis (Performance)

Can the integration of *aligned* multimodal embeddings (text, image, audio) enhance recommendation accuracy compared to unimodal or non-aligned approaches?

Pipeline


Embedding Extraction and Validation
Data Distribution Analysis
MMRec Model Training & Evaluation

Datasets Overview

Challenge: Dealing with different structural levels of multimodal data.

Feature	**MovieLens 1M**	**LastFM**
Domain	Movies	Music
Interactions	User-Movie (Explicit Rating)	User-Artist (Implicit Listen)
MM Level	**Item Level** (Movie)	**Track Level (Sub-Item)**
Modalities	Posters, Trailers, Plots	Audio, Covers, Metadata
Pipeline	Direct Mapping	**Artist-to-Track Expansion**

Table: Comparison of datasets and processing strategies

Models Evaluated

We compared performance across simpler projection models and complex Graph Neural Networks:

- **BPR (Baseline):** Bayesian Personalized Ranking (ID-only).
- **VBPR:** Visual BPR extended to support multimodal features (projection-based).
- **LATTICE:** GNN that mines latent structures between items.
- **FREEDOM:** GNN with modality-specific encoders and attention fusion.

Evaluation Protocol

Metrics Evaluated:

- **Recall@k**
- **NDCG@k**
- **Precision@k**
- **MAP@k**

Experimental Settings:

- **Cut-offs (k):** $k \in \{5, 10, 20, 50\}$
- **Filtering:** 5-core filtering applied to ensure interaction density.

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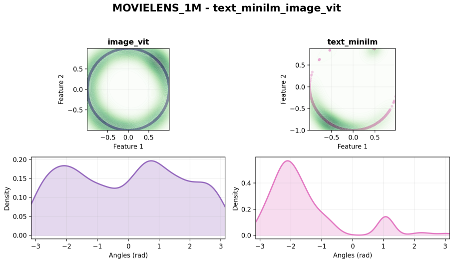
Embedding Distribution Analysis (UMAP + KDE)

To understand *why* models perform differently, we analyzed the geometric structure of the embeddings.

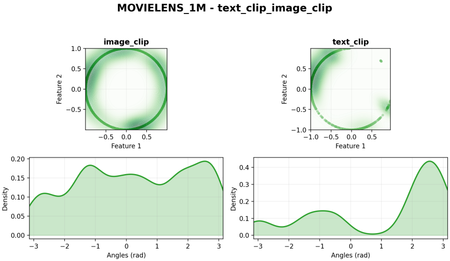
- **Technique:** UMAP projection to 2D $\rightarrow$ S1 Normalization (Unit Circle) $\rightarrow$ Gaussian KDE.
- **What to look for:**
  - *Uniformity:* Do features cover the space?
  - *Peaks:* Distinct peaks indicate specific feature orientations.
  - *Collapse:* Overlapping distributions imply loss of modality-specific signal.

MovieLens 1M: Aligned vs Disaligned

Disaligned (MiniLM+ViT)



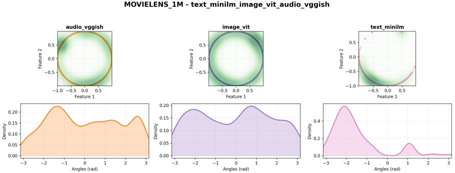
CLIP Aligned (Text+Image)



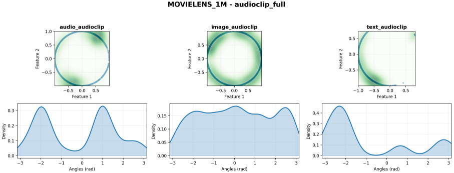
- **Disaligned:** Features maintain specific orientations, capturing different aspects of user preference (e.g., visual style vs. narrative).
- **CLIP Aligned:** Successfully brings Text and Image closer semantically while maintaining enough structural diversity to be discriminative (Best Performance).

The "AudioCLIP Anomaly"

Disaligned (Rich Signal)



AudioCLIP (Over-Aligned)



Why does alignment fail here?

AudioCLIP's aggressive alignment appears to collapse the audio feature space, making it sharply distinct from other modalities. This collapse, combined with a flattened image distribution with respect to CLIP

Quantitative Results: MovieLens 1M (k=10)

VBPR with CLIP features generally outperforms others. AudioCLIP underperforms.

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1776	0.2159	0.1631	0.1120
VBPR (CLIP)	**0.1911**	**0.2300**	**0.1745**	**0.1211**
VBPR (AudioCLIP)	0.1570	0.1922	0.1476	0.0969
VBPR (Disaligned)	0.1876	0.2271	0.1718	0.1195
LATTICE_3 (CLIP)	0.1877	0.2275	0.1721	0.1198
LATTICE_3 (Disaligned)	0.1848	0.2257	0.1710	0.1183
FREEDOM (CLIP)	0.1796	0.2211	0.1657	0.1158
FREEDOM (AudioCLIP)	0.1844	0.2259	0.1688	0.1193

Table: MovieLens 1M results. **VBPR (CLIP)** achieves state-of-the-art.

Quantitative Results: LastFM (k=10)

In music domain, disaligned features preserve distinct signals better.

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1132	0.1534	0.1404	0.1061
VBPR (CLIP)	0.1222	0.1646	0.1488	0.1149
VBPR (Disaligned)	**0.1307**	**0.1755**	**0.1560**	**0.1242**
LATTICE_3 (CLIP)	0.1232	0.1667	0.1507	0.1160
LATTICE_3 (Disaligned)	0.1258	0.1722	0.1535	0.1209
FREEDOM (Disaligned)	0.0986	0.1322	0.1224	0.0954

Table: LastFM results. **VBPR (Disaligned)** outperforms aligned models.

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Discussion: Context Matters

MovieLens 1M (Movies)

- Modalities share high semantic correlation.
- **Result:** Alignment (CLIP) helps. It reinforces the shared signal.
- **AudioCLIP Anomaly:** Loss in performance with respect to CLIP

LastFM (Music)

- Modalities do not share high semantic correlation.
- **Result:** Alignment does not help, even more if it is a strict alignment (AudioCLIP).

Conclusions and Future Work

- **Context-Dependent Alignment:** Aligned features (CLIP) help when modalities are semantically correlated (MovieLens), but can hurt when they are not (LastFM).
- **Aggressive Alignment (AudioCLIP):** Can hurts performances.
- **Simplicity Wins:** Simple models often beat complex ones.
- **GNNs models:** They work better with disaligned features, as they can learn to fuse different signals.
- **Future Directions:**
  - Investigate regularization techniques to prevent the an aggresive alignment with AudioCLIP-like models.
  - Automatic selection system for recommendation algorithms and encoders based on data distribution

Thank you!

Thank you for your attention!

🚀

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